

Progress report : Browser Proxy Extension for Dark Pattern Detection

COE 49414 - Human Privacy and Security

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1. Abstract

Dark patterns interface designs that intentionally steer users toward unwanted or less privacy-friendly decisions are increasingly common across the modern web. While past research has mapped these patterns and built large offline datasets for analysis, far less work has explored real-time, user-facing tools that can support people as they browse. In this project, we develop a lightweight Chrome extension that detects common dark-pattern categories using heuristic rules derived from the page’s DOM, styling, and visible text. The extension highlights suspicious elements and provides brief explanations to help users make more informed choices without disrupting their browsing flow. We also design a controlled, within-subjects user study to evaluate how the tool affects task performance, error rates, and perceived control. Our goal is to create a transparent, rules-first proof of concept that demonstrates whether real-time assistance can meaningfully help users recognize and resist manipulative interface designs.

2. Introduction

Dark patterns have become a routine part of online experiences, especially in areas like cookie banners, checkout flows, and subscription settings. These design choices such as hidden opt-outs, pre-selected options, and visually biased buttons often push users toward decisions that benefit the website rather than the user. Although prior work has created taxonomies, datasets, and detection frameworks, most of these efforts operate offline or require complex crawling pipelines. As a result, they provide limited support for users who encounter dark patterns in real time.

Our project addresses this gap by building a browser extension that detects dark patterns directly on the page as the user interacts with it. The system combines heuristic rules for structure, text, and styling with the option to apply a lightweight local model for ambiguous cases. Suspicious elements are highlighted and explained in a simple sidebar, giving users clearer insight into how an interface may be nudging them.

To understand the tool’s usefulness, we designed a small, human-centered evaluation with tasks commonly affected by dark patterns. Participants will complete these tasks both with and without the extension, allowing us to measure changes in accuracy, speed, and perceived control. The goal of this project is not to create a perfect detector, but rather to explore whether a transparent, real-time assistant can make users more aware of manipulative patterns and support more privacy-protective choices.

3.Related Work

Dark patterns have been widely recognized as an emerging concern in human-computer interaction and online privacy. Early work has focused on defining and taxonomizing deceptive interface practices that show how visual salience, defaults, and interface frictions can systematically nudge users toward less privacy-protective choices. More recent studies have looked at dark patterns in specific domains such as e-commerce, mobile apps, and consent dialogs and found these designs to be highly prevalent and often intentionally optimized to increase conversions rather than support informed consent.

A large body of work has focused on cookie banners and consent dialogs. The automated analysis of thousands of banners has demonstrated that apparently minor design changes, like asymmetric button sizes, color contrasts, or multi-step "reject all" flows, can strongly influence acceptance rates (Soe et al., 2022). These systems often rely on large-scale crawling and offline analysis pipelines rather than real-time support of end users while browsing (Mathur et al., 2019), which limits their usefulness as everyday privacy tools.

Other works have created dark-pattern datasets and their respective detection frameworks. Recent corpora of e-commerce interfaces, for instance, annotated urgency messages, scarcity cues, pre-ticked checkboxes, and confirm-shaming language, and assessed the performance of baseline machine-learning models-e.g., BERT-based classifiers-on those annotations (Yada et al., 2022; Gunawan et al., 2021). These works illustrate how language models can learn to catch some textual cues about manipulation, but they tend to neglect layout and styling details highly relevant for making sense of how an interface "nudges" the user.

There is also developing work on measurement-driven detection frameworks that extract features from observable HCI signals-such as click counts, scroll depth, or the number of steps to reach an opt-out-and ontologies of dark-pattern knowledge that formalize pattern categories and their relations to support shared annotation and reasoning (Kocyigit et al., 2024; Gray et al., 2024). These contributions are important conceptual and technical building blocks, but most prototypes remain research code, require substantial infrastructure, or target a single domain. Finally, a few browser extensions have appeared in public stores that attempt to warn users about dark patterns (Wood 2023a; Wood 2023b). Such tools usually focus on one subset of the patterns-for instance, tracking consent-or rely on simple keyword lists, where little or no transparency exists regarding their design and evaluation. To the best of our knowledge, there is still a gap for a lightweight, rules-first Chrome extension that could combine DOM structure, styling, and textual cues and optionally incorporate a small on-device model for ambiguous cases along with being explicitly designed to be evaluated through a controlled user study. This is what our current project aims to fill.

Taxonomy of Dark Patterns

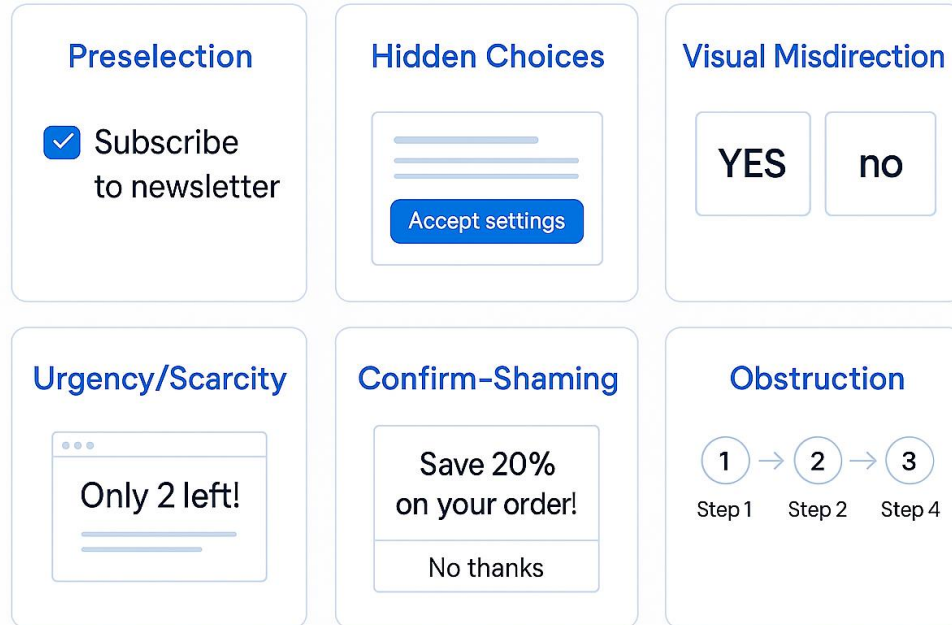


Figure 1 : Taxonomy of Dark Patterns

4. Proposed Method

4.1 System Overview

We present a Chrome extension performing real-time dark-pattern detection in a fully in-browser fashion. Once a page finishes loading, a content script examines the DOM, visible text, and selected style attributes in order to identify the interface elements which are likely to implement dark patterns, highlighting suspicious elements and providing concise explanations along with suggested privacy-friendly actions through a side panel. The system is designed with the following principles in mind: transparent (the rules and heuristics are inspectable), lightweight (no tracking or logging of the user's personal browsing history outside the user's control), and Manifest V3-compatible.

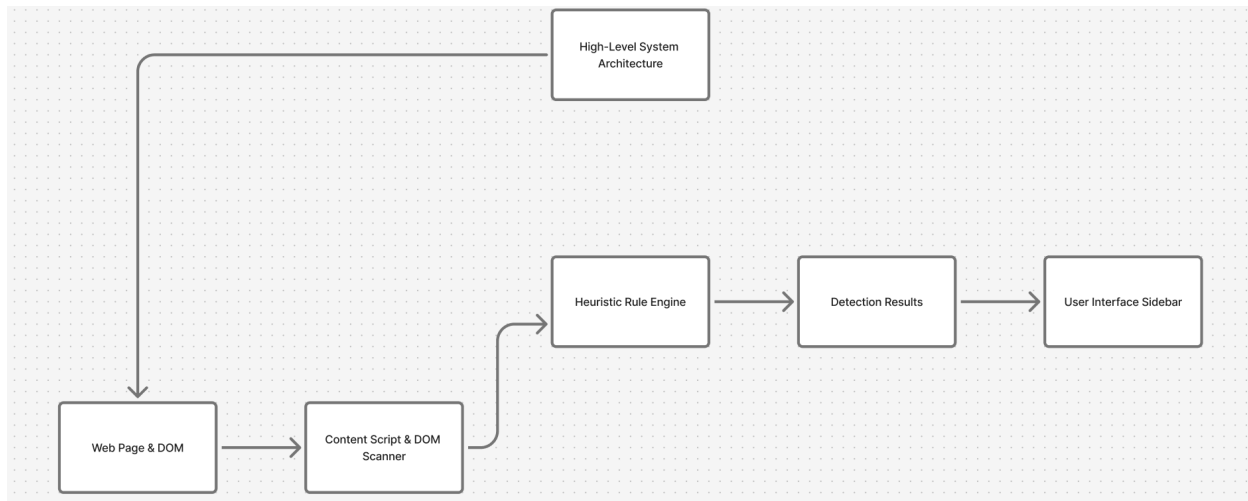


Figure 2: High-Level System Architecture

4.2 Dark-Pattern Categories and Heuristic Rules

We focus on six dark-pattern categories that are both common in practice and amenable to rule-based detection:

1. Preselection – options that are pre-ticked or pre-selected in favor of more intrusive data practices (e.g., marketing email checkboxes).
2. Hidden Choices – privacy-protective options that require extra clicks, are placed on secondary screens, or are visually de-emphasized.
3. Visual Misdirection – asymmetric button sizes, colors, or positions that strongly favor data-sharing actions.
4. Urgency/Scarcity – time-limited offers or artificial stock warnings (e.g., “Only 2 left!”) that pressure users into quick decisions.
5. Confirm-Shaming – emotionally loaded phrasing that frames refusal as irresponsible or undesirable.
6. Obstruction / Forced Action – flows that intentionally increase friction for opting out (e.g., multiple confirmation dialogs or long, mandatory forms).

For each category, we define a set of heuristic cues that can be computed from the DOM and CSS:

- For preselection, the extension inspects input elements (checkboxes, radio buttons, toggles) with checked attributes, combined with nearby labels that contain marketing- or tracking-related keywords.
- For hidden choices and obstruction, it counts the number of clicks needed to reach “reject all” or unsubscribe actions and flags flows where the opt-out path is significantly longer than the accept path.
- For visual misdirection, it compares button size, color contrast, and position between primary (e.g., “Accept all”) and secondary (e.g., “Manage settings”) actions to identify strong asymmetries.
- For urgency/scarcity and confirm-shaming, regular expressions match characteristic textual patterns such as countdown timers, stock messages, or shaming phrases (e.g., “don’t miss out”, “I don’t care about privacy”).

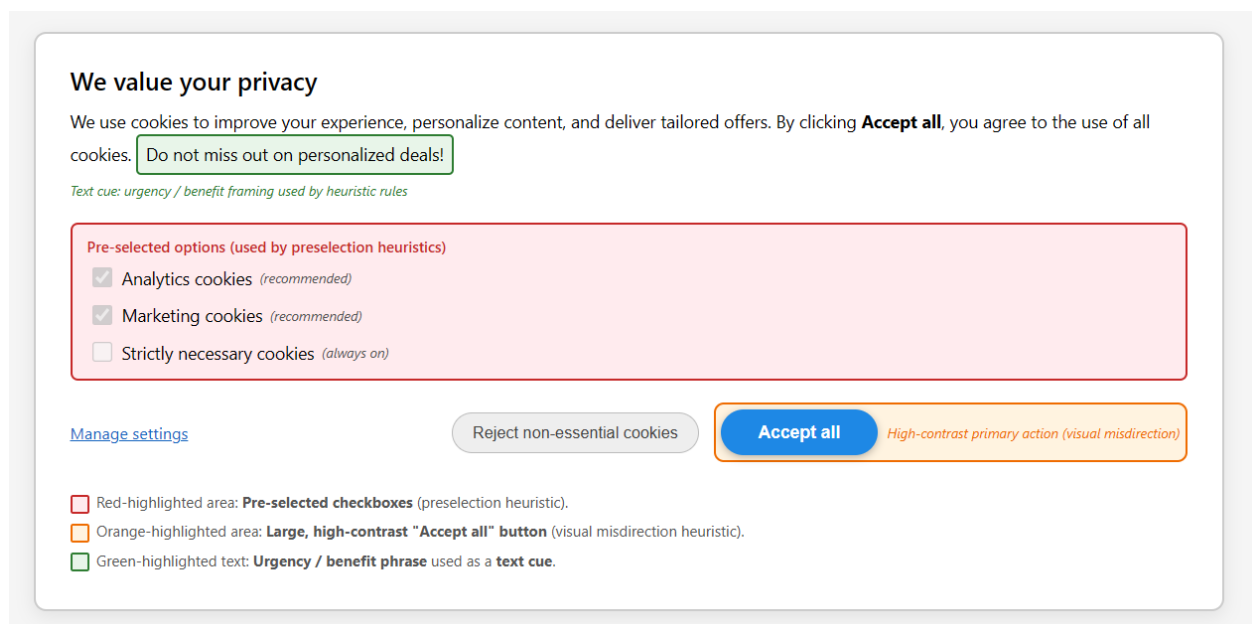


Figure 3: Example DOM Snippet with Highlighted Heuristic Features.

4.3 DOM and Text Analysis Pipeline

When a page finishes loading, a MutationObserver-based scanner traverses key interface regions such as modal dialogs, banners, forms, and checkout steps. The pipeline proceeds as follows:

1. **Element Collection.** The content script collects candidate elements (buttons, inputs, dialogs, banners) based on tag names and ARIA roles.

2. **Feature Extraction.** For each candidate, the system extracts structural features (DOM depth, sibling relationships), visual features (bounding box size, font weight, computed styles), and textual content (inner text plus, optionally, text recognized from images via Tesseract.js).
3. **Rule Evaluation.** The heuristic engine evaluates category-specific rules on these features. Each rule contributes a score; scores are aggregated to yield a per-element suspicion level and a category label.
4. **Optional Model Inference.** If the user enables it, a small on-device text classifier (e.g., TF.js or ONNX Runtime Web) is applied to borderline cases to refine category assignments, especially for nuanced confirm-shaming or urgency language.
5. **Result Aggregation and Filtering.** Overlapping detections are merged, and low-confidence hits are discarded to reduce noise.

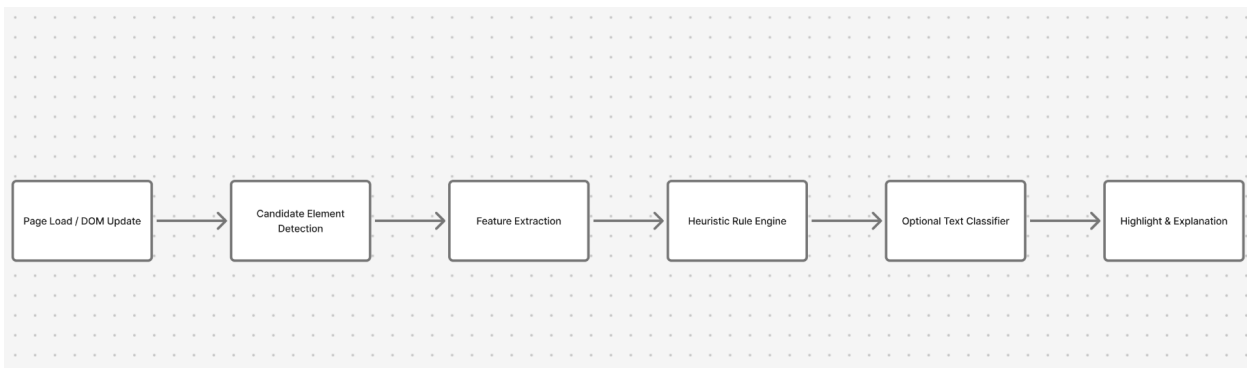


Figure 4: Detection Pipeline Flowchart.

The extension communicates its findings via a minimal, non-intrusive UI:

- Suspicious elements are outlined with subtle color or icon overlays to indicate their detected category.
- Clicking the overlay opens a sidebar that summarizes which pattern was detected, why it might be manipulative, and what a safer alternative action might be. That could be a direct link to “Reject non-essential cookies” or instructions to untick a checkbox.
- It allows users to toggle detection categories on or off, adjust sensitivity thresholds, and temporarily disable the extension for certain sites.

This design supports informed decision-making without overloading users with warnings; it also aligns with the planned user study in which participants will perform browsing tasks both with and without the extension, while measuring changes in privacy-protective choices, task time, and perceived control.

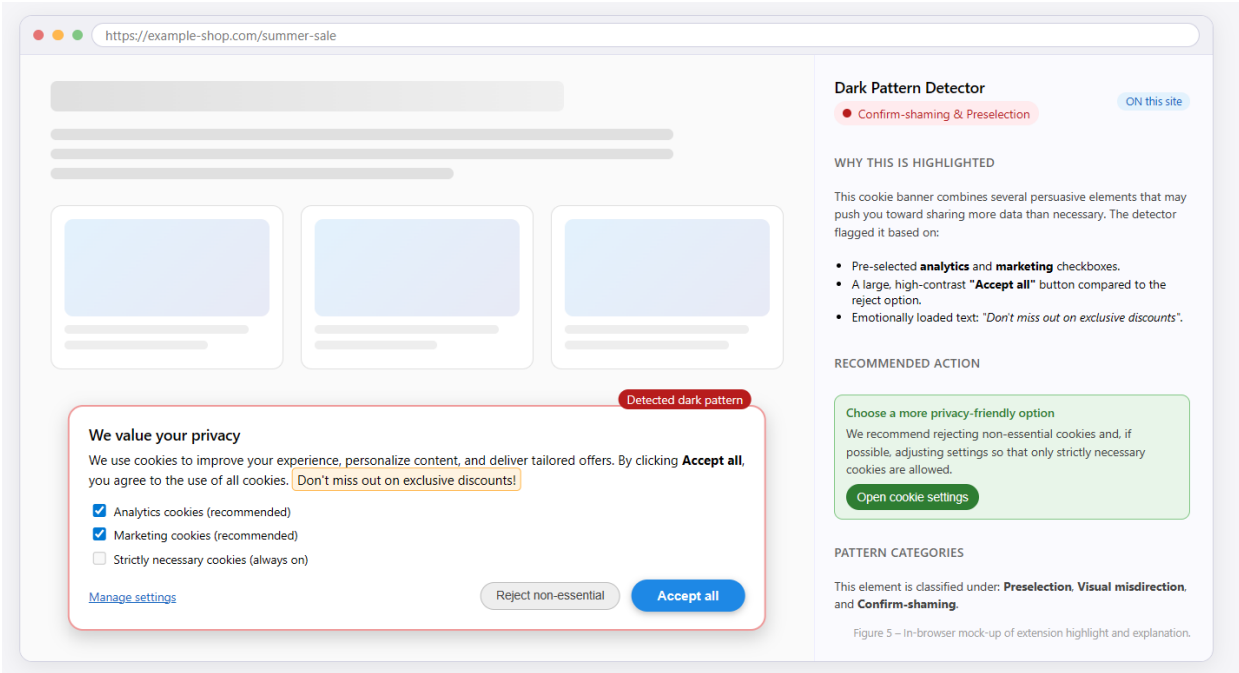


Figure 5: In-Browser UI Mock-Up

5. Plan for Evaluation (User Study Design)

This evaluation plan follows a human-centered, experimental approach that aligns with user study design, variable selection, and human-in-the-loop reasoning. It builds directly on the objectives outlined in our Project Proposal for the browser extension that detects dark patterns in real time.

5.1 Evaluation Goals

Our evaluation consists of two main components:

1. **Technical Evaluation**
Measuring how accurately the extension identifies each of the six dark pattern categories defined in the proposal (precision, recall, F1).
2. **Human-Centered Evaluation**
Determining whether the extension improves users' ability to identify and avoid deceptive design, reduces errors in privacy decisions, and increases feelings of control and trust.

This dual approach reflects the whole-system view covered in class, where both technical performance and human behavior must be considered.

5.2 Research Questions

- RQ1:** Does the extension improve users' ability to identify and avoid dark patterns?
- RQ2:** Does the extension reduce user errors and increase privacy-protective actions?
- RQ3:** How do users perceive the extension in terms of usability, trust, cognitive effort, and control?
- RQ4:** Does the UI effectively support attention, comprehension, and retention without overloading users?
- RQ5:** How accurate is the detection pipeline on labeled pages, and which categories perform best?

5.3 Study Type

We adopt a **within-subjects experimental design**, where each participant completes tasks in two conditions:

- **Extension OFF**
- **Extension ON**

This design reduces variance caused by individual differences

Counterbalancing

To eliminate order effects, participants are split into two balanced groups:

- Group A: OFF → ON
- Group B: ON → OFF

This minimizes learning, fatigue, and sequence bias.

5.4 Study Variables

Independent Variable (IV)

Factor	Levels
Extension Condition	OFF, ON

Dependent Variables (DVs)

Behavioral Measures

- Task completion time
- Number of clicks
- Error rate
- Task success (binary)

Subjective Measures

- NASA-TLX (effort)
- Trust and perceived control (Likert)
- System Usability Scale (SUS)

Qualitative Measures

- Interview feedback on clarity, false positives, UI comprehension, and cognitive load

Control Variables

- Browser (Chrome desktop)
- Device type and screen size
- Lighting and testing environment
- Identical task order within each condition

Random Variables

- Participant technical familiarity
- Prior experience with dark patterns
- General browsing skills

These help maintain external and ecological validity.

5.5 Task Design

Tasks represent real interactions where dark patterns appear:

1. Rejecting non-essential cookies in misleading banners
2. Disabling pre-selected marketing checkboxes during checkout
3. Locating genuine opt-out options in subscription settings
4. Identifying urgency and scarcity cues (e.g., “Only 2 left”)
5. Navigating deceptive consent modals

These tasks are structured to (1) reflect real-life behavior and (2) discriminate clearly between ON/OFF conditions, which is consistent with good task design principles from class.

5.6 Study Procedure

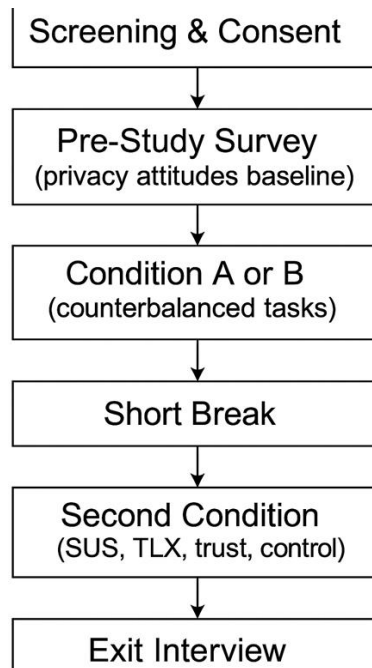


Figure 6: Study Procedures

5.7 Participant Plan

- **Sample Size:** 24–30 participants
- **Recruitment:** Convenience sampling through friends, family, and classmates
- **Inclusion Criteria:** Regular web users
- **Exclusion:** Anyone involved in building the extension
- **Compensation:** Optional small token (if needed), but not required

5.8 Ethical Considerations

This is a **course project**, not formal human-subject research, so **IRB approval is not required**. Instead, we will ensure basic ethical handling of participants:

- Participants will be recruited informally (friends, family, classmates)
- No personal browsing history or sensitive data will be collected
- Participation is voluntary, and users may stop at any time
- Only task performance metrics and survey responses will be recorded
- All data will remain anonymous and used solely for this class project

These steps ensure the study remains low-risk and respectful while avoiding unnecessary administrative procedures.

5.9 Analysis Plan

Quantitative Analysis

Variable Type	Statistical Test	Notes
Completion Time	Paired t-test or Wilcoxon	Depends on normality
Success Rate	McNemar's test	Binary paired
Clicks / Errors	Wilcoxon	Non-normal expected
Likert Ratings	Wilcoxon	Ordinal
SUS Score	Descriptive + comparison	Standard scoring

Effect sizes (Cohen's d or r) will be included.

Detection Accuracy

Using the labeled dataset:

- Precision, recall, F1 per dark pattern category
- Confusion matrices to analyze common errors
- Comparison between “rules only” and “rules + optional tiny model”

Qualitative Analysis

- Semi-structured interviews analyzed via thematic analysis
- Hybrid coding (inductive + deductive)
- Partial double coding for reliability

5.10 Expected Findings

We expect to observe:

- Higher success rates and fewer errors when the extension is active
- Reduced time and fewer clicks to reach privacy-protective outcomes

- Higher perceived control and trust
- Positive SUS ratings indicating usability
- Some false positives, particularly in ambiguous urgency patterns
- Stronger precision in DOM-based cues than purely textual cues

These findings will guide future refinement of detection signals and UI explanations.

5. Current Progress and Remaining Tasks

Completed So Far:

- Chosen the dark-pattern categories and signals
- Started early DOM/text detection ideas
- Set up basic scanning and regex structure
- Gathered a small set of example pages for testing
- Finished the study design and evaluation plan

Remaining Tasks

1. Continue building and finishing the extension code
2. Finalize detection heuristics and thresholds
3. Label more pages for the evaluation dataset
4. Finish the UI (highlighting + simple explanations)
5. Do an internal pilot test
6. Recruit friends, family, and classmates for the study
7. Analyze the results
8. Prepare the final write-up and presentation

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